

Dataset Identification:-

- Dataset: Facebook Social Circles (SNAP)
- Source File: facebook_edges.csv
- 4,039 Total Nodes
- 88,234 Total Edges

Key Network Influencers:-

- Centrality analysis was performed to identify highly connected hubs and influential bridges within the network.

Rank	Top 5 by Degree (Popularity)	Top 5 by Betweenness (Influence)
1	Node 107 (1045)	Node 107
2	Node 1684 (792)	Node 1684
3	Node 1912 (755)	Node 1912
4	Node 3437 (547)	Node 1085
5	Node 0 (347)	Node 0

Visualizations:-

Id	Label	Interval	Modularity Class	Degree
25			1	522
30			2	511
27			2	484
28			6	459
0			1	451
32			7	448
23			0	427
29			1	419
31			5	419
26			7	417
24			6	389
41			3	357
39			4	344
49			3	340
35			1	333
...			...	...

Id	Label	Interval	Modularity Class	Degree	Eccentricity	Closeness Centrality	Harmonic Closeness Centrality	Betweenness Centrality
25			1	522	3.0	0.534198	0.56443	0.025157
30			2	511	3.0	0.533351	0.563026	0.021335
27			2	484	3.0	0.531736	0.559848	0.020175
28			6	459	3.0	0.529782	0.556629	0.019539
0			1	451	3.0	0.529504	0.555803	0.01864
32			7	448	3.0	0.529019	0.555267	0.017312
29			1	419	3.0	0.526741	0.551511	0.016982
23			0	427	3.0	0.527774	0.55279	0.016929
31			5	419	3.0	0.527016	0.551676	0.016137
26			7	417	3.0	0.526535	0.551222	0.014845
24			6	389	3.0	0.524347	0.54759	0.012416
41			3	357	3.0	0.520696	0.542719	0.011562
34			6	331	3.0	0.520093	0.540201	0.011293
49			3	340	3.0	0.520026	0.540903	0.011207
39			4	344	3.0	0.520965	0.541811	0.010596
35			1	333	3.0	0.519424	0.539954	0.010394
9			2	326	3.0	0.51849	0.538798	0.009687
36			0	320	3.0	0.518623	0.538385	0.009593
58			1	320	3.0	0.517427	0.537642	0.009068
37			0	299	3.0	0.516699	0.535455	0.008036
33			6	296	3.0	0.516097	0.534331	0.008036

Interpretation of Results:-

- The network exhibits a clear "Small-World" structure. The presence of large hubs (like Node 107) suggests that a few individuals hold significant power in connecting disparate groups. The high clustering coefficient observed in the community visualization indicates that users are likely grouped into real-world social circles (colleagues, students, etc.).
- Betweenness centrality identified Node 1085 as a key "bridge." While this user doesn't have the highest number of friends, they are essential for information flow between different clusters. Removing such bridges would fragment the network more severely than removing average nodes.

Lessons Learned:-

- Structural Importance: Learned that a node's value isn't just about direct connections (Degree) but its position in the broader path (Betweenness).
- Tool Proficiency: Gained experience using Gephi for spatial layouts and modularity calculations to reveal hidden social patterns.
- Data Narratives: Discovered how to translate complex CSV edge-lists into visual stories that explain human interaction.